3.5.8 Definite Integrals

`Integrate[`f`, `x`]`	the indefinite integral $\int f\,dx$
`Integrate[`f`, {`x`, `$xmin$`, `$xmax$`}]`	the definite integral $\int_{xmin}^{xmax} f\,dx$
`Integrate[`f`, {`x`, `$xmin$`, `$xmax$`}, {`y`, `$ymin$`, `$ymax$`}]`	the multiple integral $\int_{xmin}^{xmax} dx\ \int_{ymin}^{ymax} dy\ f$

Integration functions.

Here is the integral $\int_a^b x^2\,dx$.

```
In[1]:= Integrate[x^2, {x, a, b}]

           3    3
         -a    b
Out[1]= --- + --
          3    3
```

This gives the multiple integral $\int_0^a dx\,\int_0^b dy\,(x^2+y^2)$.

```
In[2]:= Integrate[x^2 + y^2, {x, 0, a}, {y, 0, b}]

         3         3
        a  b    a b
Out[2]= ---- + ----
         3       3
```

The `y` integral is done first. Its limits can depend on the value of `x`. This ordering is the same as is used in functions like `Sum` and `Table`.

```
In[3]:= Integrate[x^2 + y^2, {x, 0, a}, {y, 0, x}]

         4
        a
Out[3]= --
        3
```

You can often do a definite integral by first finding the indefinite one, and then explicitly substituting in the limits. You have to be careful, however, when the integration region contains a singularity. The integral $\int_{-1}^{1} x^{-2}\,dx$, for example, has an indefinite form which is finite at each end point. Nevertheless, the integrand has a double pole at $x = 0$, and the true integral is infinite.

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